

Special Problem Defense

MitoChime

A Machine Learning Pipeline for Detecting PCR-Induced
Chimeric Reads in Mitochondrial Illumina Data

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Opening question: which read is artificial?

Opening question: which read appears artificially joined?

Read A



continuous alignment pattern

Read B



possible junction

The task of MitoChime is to automate this decision at read scale.

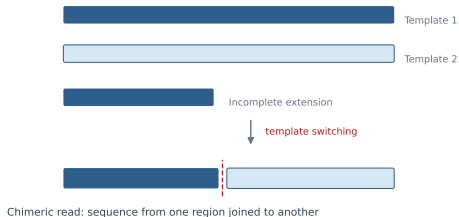
Framing question

Which read shows evidence of being artificially joined?

- ▶ Sequencing datasets contain thousands of reads.
- ▶ Manual inspection is not practical.
- ▶ The goal is automated read-level detection.

Real-world problem

PCR-induced chimera formation



PCR amplification can produce **artificial hybrid reads**.

These reads may be included in sequencing data and mislead mitochondrial genome assembly.

Core issue

Some reads appear valid but do not represent the original mitochondrial sequence.

Why the problem matters

Assembly

Fragmentation

Contigs may break or branch

Analysis

False signals

Incorrect junctions may appear

Interpretation

Lower reliability

Downstream conclusions weaken

For *Sardinella lemuru*, unreliable mitochondrial assemblies may affect fisheries genetics, population structure, and evolutionary analysis.

Existing approaches

Reference-based

Compares query sequences against known non-chimeric references.

De novo

Uses internal dataset patterns such as abundance and similarity.

Tool-based

Includes UCHIME, UCHIME2, DADA2, CATCh, and ChimPipe.

Existing tools and the study gap

	Amplicon	ASV/OTU	Read-level	Mito PCR chimera
UCHIME	Yes	Yes	Limited	No
DADA2	Yes	Yes	No	No
CATCh	Yes	Yes	No	No
ChimPipe	No	No	Yes	Different
MitoChime	No	No	Yes	Yes

Most existing tools were designed for:

- ▶ microbial amplicon or ASV/OTU-level analysis,
- ▶ reference-based or abundance-based detection,
- ▶ community-level filtering,
- ▶ biological chimeras or other sequencing contexts.

Gap addressed by this study

There is limited support for direct **read-level detection** of PCR-induced chimeras in single-species mitochondrial Illumina data.

Can machine learning detect PCR-induced chimeric reads
before mitochondrial genome assembly?

Input

Simulated clean and chimeric paired-end
Illumina reads from *S. lemuru*

Output

A modular pipeline that predicts and filters
suspected chimeric reads

Scope and limitations

Included

- ▶ *Sardinella lemuru*
- ▶ mitochondrial Illumina reads
- ▶ PCR-induced chimeras
- ▶ simulated labelled datasets

Excluded

- ▶ long-read sequencing
- ▶ NUMTs
- ▶ naturally occurring chimeras
- ▶ broad multi-species validation

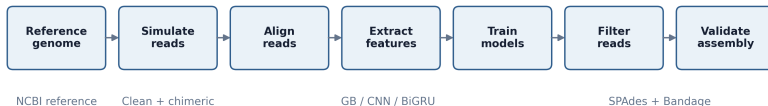
The study prioritized a focused and controlled evaluation setting.

Methodology

From simulated reads to model-based filtering

MitoChime workflow

MitoChime workflow



Overview

The workflow generated labelled reads, extracted read-level evidence, trained models, filtered suspected chimeras, and evaluated downstream assembly behavior.

Data construction

Clean reads

Simulated from the *S. lemuru* mitochondrial reference genome.

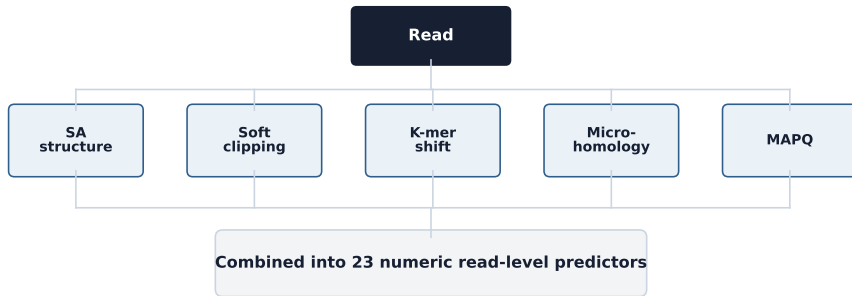
Chimeric reads

Generated by joining non-adjacent mitochondrial segments to mimic PCR template switching.

Labels: **clean** = 0 and **chimeric** = 1

Feature extraction

Read-level feature groups



Feature representation

Each read was converted into a set of 23 numeric predictors describing alignment structure, clipping behavior, sequence composition, microhomology, and mapping quality.

Dataset representation

The labelled reads were prepared in two dataset formats.

read_id	label	seq
NC_039553.1_10001_10492_5:0:0_8:0:0_b77/1	0	GTAAGGCATAGTAAGGCAATGGTGAGGCTGTGTGTGATTGCA
NC_039553.1_10001_10492_5:0:0_8:0:0_b77/2	0	GAGTGAGCCGAATAGGGAATTAGTCCAACCTAAGACTCCTGA
NC_039553.1_10003_10548_5:0:0_8:0:0_f4d/1	0	GTGAGCCGAAAAGGTAATTAGTCCAACCTACGACTCCTGATT
NC_039553.1_10003_10548_5:0:0_8:0:0_f4d/2	0	TGGCTATATAGTGTTGGTGAGGGCTCAGGCCGTTTCCGTTTC
NC_039553.1_10005_10512_6:0:0_6:0:0_c35/1	0	CCGTTGTCCAGTTTAGCCAGGTAAGGCTTAGTAAGGCAATGC
NC_039553.1_10005_10512_6:0:0_6:0:0_c35/2	0	GAGCCGCATAGGGCAATAGTCCAACCTAAGACTCGTGATTTC
NC_039553.1_10008_10523_3:0:0_10:0:0_914/1	0	TCAGCCCCTTTCCGTTGACCAGTTTAGCCAGGTAAGGCTTAC
NC_039553.1_10008_10523_3:0:0_10:0:0_914/2	0	CCGAATAGGGACATAGTCCAACCTAAGACTCCTGATTTCCTGC
NC_039553.1_10012_10513_12:0:0_5:0:0_1340/1	0	TCCGTTGTGCAGTTTAGCCAGGTAAGGCTTAGTAAGGCAATG
NC_039553.1_10012_10513_12:0:0_5:0:0_1340/2	0	ATAGTGAATTAGTCCAAGTTAAGACTCCTGATTTCGGCTCAGA
NC_039553.1_10012_10591_3:0:0_5:0:0_2571/1	0	AGCCAGCATGTAAGAACCACAGCGGGGAGGATCAGGGATC
NC_039553.1_10012_10591_3:0:0_5:0:0_2571/2	0	ATATGGAATTAGTCCAACCTAAGACTCCTGATTTCGGCTCAGA
NC_039553.1_10015_10459_2:0:0_8:0:0_253e/1	0	GTGATTGCAGCAGTTCAAAGTGACTTTTATAGCGCGAAATCAG
NC_039553.1_10015_10459_2:0:0_8:0:0_253e/2	0	GGGAATTAGTCCAACCTAAGAGTCTGATTTCGGCTCAGATG.
NC_039553.1_10015_10585_10:0:0_11:0:0_d4c/1	0	GGGAATTAGACGAACATAATCTCCTGAATTCGGCTCAGATG/
NC_039553.1_10015_10585_10:0:0_11:0:0_d4c/2	0	CATGTAAGAACCCACAGCGGGGAGGCTAAGGGATCCCTGGC

read_id	label	read_len	mean_bp	ref_name	ref_str	strand	mapq	cigar	has_si	sa_count	num_seg	sa_diff	config	sa_min_delta_pos	sa_max_delta_pos
NC_039553.1_3_540_8:0:0_6:0:0_a12	0	100	13.000	NC_039553.1	3	0	60	150M	0	0	1	0		0	0
NC_039553.1_4_430_13:0:0_11:0:0_243d	0	100	13.000	NC_039553.1	4	0	60	150M	0	0	1	0		0	0
NC_039553.1_5_405_8:0:0_11:0:0_1440	0	100	13.000	NC_039553.1	5	0	60	150M	0	0	1	0		0	0
NC_039553.1_6_523_8:0:0_9:0:0_82u	0	100	13.000	NC_039553.1	6	0	60	150M	0	0	1	0		0	0
NC_039553.1_9_574_7:0:0_7:0:0_181b	0	100	13.000	NC_039553.1	9	0	60	150M	0	0	1	0		0	0
NC_039553.1_10_501_9:0:0_8:0:0_256b	0	100	13.000	NC_039553.1	10	0	60	150M	0	0	1	0		0	0
NC_039553.1_11_509_8:0:0_11:0:0_a19	0	100	13.000	NC_039553.1	11	0	60	150M	0	0	1	0		0	0
NC_039553.1_12_427_9:0:0_9:0:0_157	0	100	13.000	NC_039553.1	12	0	60	150M	0	0	1	0		0	0
NC_039553.1_12_489_8:0:0_9:0:0_86f	0	100	13.000	NC_039553.1	12	0	60	150M	0	0	1	0		0	0
NC_039553.1_12_593_13:0:0_13:0:0_2375	0	100	13.000	NC_039553.1	12	0	24	150M	0	0	1	0		0	0
NC_039553.1_14_515_7:0:0_7:0:0_20a4	0	100	13.000	NC_039553.1	14	0	60	150M	0	0	1	0		0	0
NC_039553.1_15_525_9:0:0_9:0:0_1e7e	0	100	13.000	NC_039553.1	15	0	60	150M	0	0	1	0		0	0
NC_039553.1_17_520_4:0:0_8:0:0_25c	0	100	13.000	NC_039553.1	17	0	60	148MM	0	0	1	0		0	0
NC_039553.1_18_522_8:0:0_9:0:0_c84	0	100	13.000	NC_039553.1	18	0	60	150M	0	0	1	0		0	0
NC_039553.1_18_593_8:0:0_3:0:0_sca	0	100	13.000	NC_039553.1	18	0	60	150M	0	0	1	0		0	0
NC_039553.1_18_482_8:0:0_7:0:0_1e4b	0	100	13.000	NC_039553.1	18	0	60	150M	0	0	1	0		0	0
NC_039553.1_19_516_8:0:0_8:0:0_c1a	0	100	13.000	NC_039553.1	19	0	60	150M	0	0	1	0		0	0
NC_039553.1_20_525_9:0:0_10:0:0_1b37	0	100	13.000	NC_039553.1	20	0	60	150M	0	0	1	0		0	0
NC_039553.1_21_560_10:0:0_9:0:0_189b	0	100	13.000	NC_039553.1	21	0	60	150M	0	0	1	0		0	0
NC_039553.1_23_443_2:0:0_8:0:0_1009	0	100	13.000	NC_039553.1	23	0	60	150M	0	0	1	0		0	0
NC_039553.1_25_571_7:0:0_8:0:0_1932	0	100	13.000	NC_039553.1	25	0	60	150M	0	0	1	0		0	0
NC_039553.1_28_536_8:0:0_8:0:0_182	0	100	13.000	NC_039553.1	28	0	60	150M	0	0	1	0		0	0
NC_039553.1_32_487_8:0:0_8:0:0_2005	0	100	13.000	NC_039553.1	32	0	60	150M	0	0	1	0		0	0
NC_039553.1_34_520_8:0:0_9:0:0_691	0	100	13.000	NC_039553.1	34	0	60	150M	0	0	1	0		0	0
NC_039553.1_34_452_2:0:0_16:0:0_495	0	100	13.000	NC_039553.1	34	0	60	150M	0	0	1	0		0	0
NC_039553.1_36_574_7:0:0_8:0:0_993c	0	100	13.000	NC_039553.1	36	0	60	150M	0	0	1	0		0	0

Feature matrix

For machine learning models

Sequence dataset

For sequence-based deep learning models

Models evaluated

Gradient Boosting

Feature-based model using engineered alignment and sequence-derived predictors.

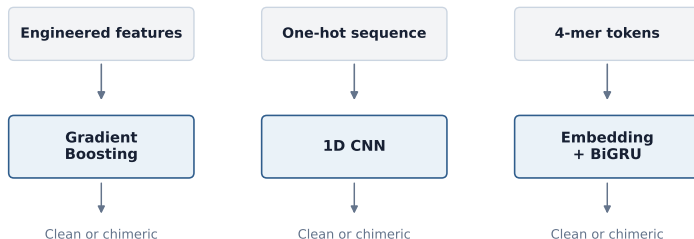
1D CNN

Sequence-based model using one-hot encoded nucleotide reads.

BiGRU

Sequence-based model using ordered k-mer tokens.

Three model representations



Evaluation strategy

Read-level evaluation

- ▶ Accuracy
- ▶ Precision
- ▶ Recall
- ▶ F1-score
- ▶ ROC-AUC

Assembly-level validation

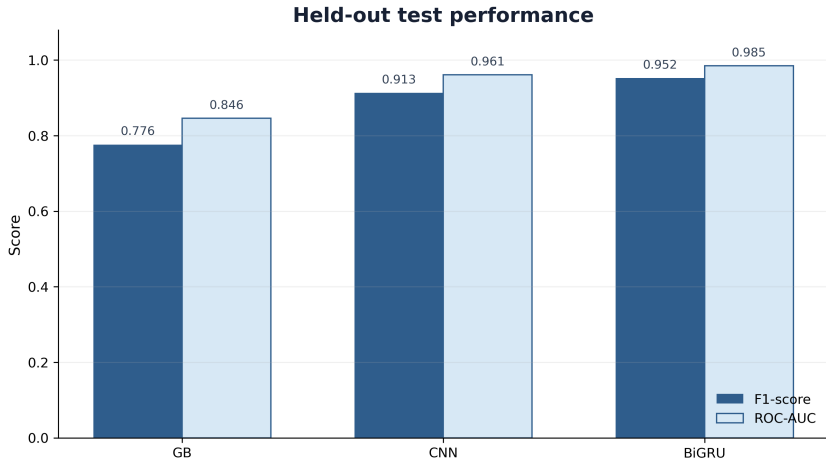
- ▶ Read retention
- ▶ Number of nodes
- ▶ Assembly graph structure
- ▶ Contiguity after filtering

The study evaluated both prediction quality and downstream utility.

Results and Discussion

Classification performance and assembly behavior

Classification performance

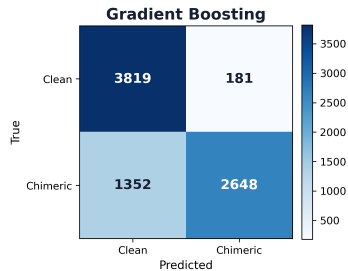


Classification performance

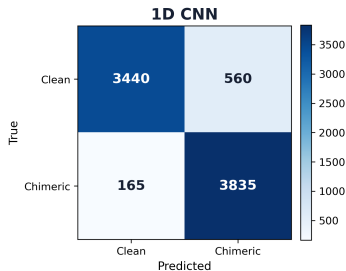
Model	F1-score	ROC-AUC	Runtime
Gradient Boosting	0.7765	0.8459	0.50 min
1D CNN	0.9135	0.9607	1.39 min
BiGRU	0.9523	0.9849	5.51 min

- ▶ **Gradient Boosting** — Conservative filtering; higher read retention
- ▶ **CNN** — Sensitive model; more aggressive filtering
- ▶ **BiGRU** — Highest F1 score; best read-level performance

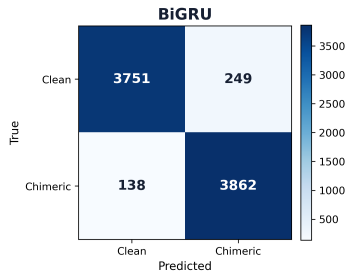
Error patterns



Gradient Boosting

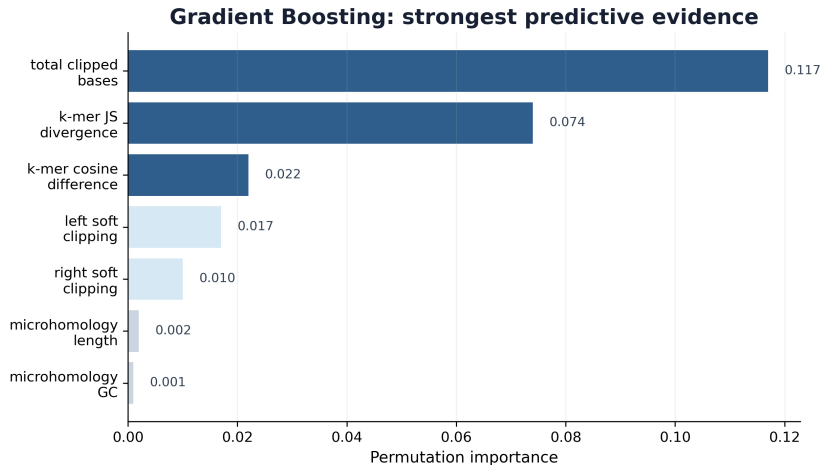


CNN



BiGRU

Feature importance



Most informative predictors: **total clipped bases**, **k-mer Jensen–Shannon divergence**, **k-mer cosine difference**, and **left soft clipping**.

Feature Family Importance

Family	Importance Sum	Importance Mean	nFeatures
Clipping	0.133830	0.033458	4
K-mer jump	0.096264	0.048132	2
Other	0.003676	0.001225	3
Microhomology	0.000347	0.000174	2
SA structure	-0.000859	-0.000066	13

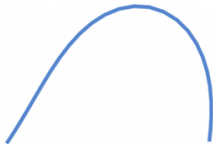
Assembly validation: high coverage



(a) Unfiltered



(b) GB



(c) CNN



(d) BiGRU

20K reads, 50% chimeras

- ▶ Unfiltered assembly showed severe graph complexity.
- ▶ Filtering simplified the graph.
- ▶ CNN produced the simplest graph.
- ▶ GB retained more reads.

Assembly validation: low coverage

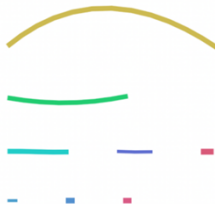


(a) Unfiltered

(b) GB



(c) CNN



(d) BiGRU

3,200 reads, 50% chimeras

At low read depth, aggressive filtering may remove too much read support and cause fragmentation.

Practical implication

The safest model depends on coverage and chimera proportion.

Recommendations: Practical Use

- ▶ Use **Gradient Boosting** when read depth is low.
- ▶ Use **BiGRU** when read-level classification is the priority.
- ▶ **CNN** may be useful in high-coverage and high-chimera conditions.

Recommendations: Future Work

- ▶ Test on real mitochondrial datasets.
- ▶ Evaluate realistic class imbalance.
- ▶ Add varied sequencing-error profiles.
- ▶ Explore hybrid feature-sequence models.

Summary

- ① PCR-induced chimeras can disrupt mitochondrial genome assembly.
- ② Existing tools do not directly address this read-level setting.
- ③ MitoChime detects suspected chimeric reads before assembly.
- ④ BiGRU achieved the strongest classification performance.
- ⑤ Model choice should consider both accuracy and read retention.

MitoChime

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Thank you.